ZERO TRUST DESIGN CLINIC

ZTA Design Clinic Activity Document

Your team's brief for the Zero Trust Access design clinic: the customer, the environment, your roles, and the five use cases to design. Capture your answers and diagrams in the **Workbook**.

Overview

This Zero Trust design clinic was created to educate the field technical resources on how Cisco's Zero Trust architecture can help organizations strengthen enterprise security while preparing for the rise of an agentic workforce. Attendees will learn how to guide meaningful customer conversations, lead effective design discussions, and identify practical opportunities to apply Zero Trust concepts across the enterprise.

In the TechElevate Zero Trust Design Clinic, attendees will have heard how Zero Trust principals apply to the One Cisco architecture and will be given a chance to apply concepts and principles into real-world customer use cases.

What you'll be able to do

By the end of this 60-minute activity you will be able to:

- Map real customer pain points and requirements to Cisco Zero Trust capabilities.
- Select the right Cisco products for each use case and justify each choice with a differentiator.
- Produce a Zero Trust design (segmentation, secure access, AI controls) shown in a network diagram.
- Explain and defend your design to a technical peer, as you would to a customer.

Design Clinic Exercise

Welcome to this design clinic. In today's complex threat landscape, static network boundaries are no longer sufficient. To achieve a true Zero Trust posture, we must shift from reactive security to an iterative, dynamic approach.

This exercise challenges your group to design a Zero Trust environment to satisfy the following 5 use cases (3 foundational and 2 optional if time permits) using what you already know from Cisco and the new agentic AI topics taught in today's session. Work together as a group and each person should assume a role below to contribute towards your final proposal.

Suggested 60-minute timeline

TIME	PHASE	WHAT TO DO
0:00-0:05	**Set up**	Assign roles; skim the customer architecture and the use cases.
0:05-0:17	**Use Case 1 · required**	Design segmentation. Start your diagram; capture your answers.
0:17-0:29	**Use Case 2 · required**	Design Zero Trust Access for remote workers; extend your diagram.
0:29-0:41	**Use Case 3 · required**	Design AI access & agent controls; extend your diagram.

TIME	PHASE	WHAT TO DO
0:41-0:52	**Present & score**	Walk your proctor through your design and reasoning; get feedback.
0:52-1:00	**Bonus if ahead**	Attempt Use Case 4 and/or 5, or refine your diagram.

How to work in the time you have:

- Use cases 1-3 are the deliverable; 4-5 are stretch goals if you finish early.
- Keep one diagram and grow it use case to use case, don't start over each time.
- Answer in short bullet points, not paragraphs. Name the product, say what it does, say why.
- Ask your proctor for a 60-second sanity check after Use Case 1 so you're on the right track.

Roles in your group

Each person in your group will assume a role.

NetOps

- Responsible for network architecture and Operations
- Often includes desktop support

Information Security

- Responsible for information security and corporate security policy
- Responsible for IAM: AD, Entra, Sailpoint, etc.

Compliance

- Validate decisions made by the architects
- Validate policies align to customer requirements

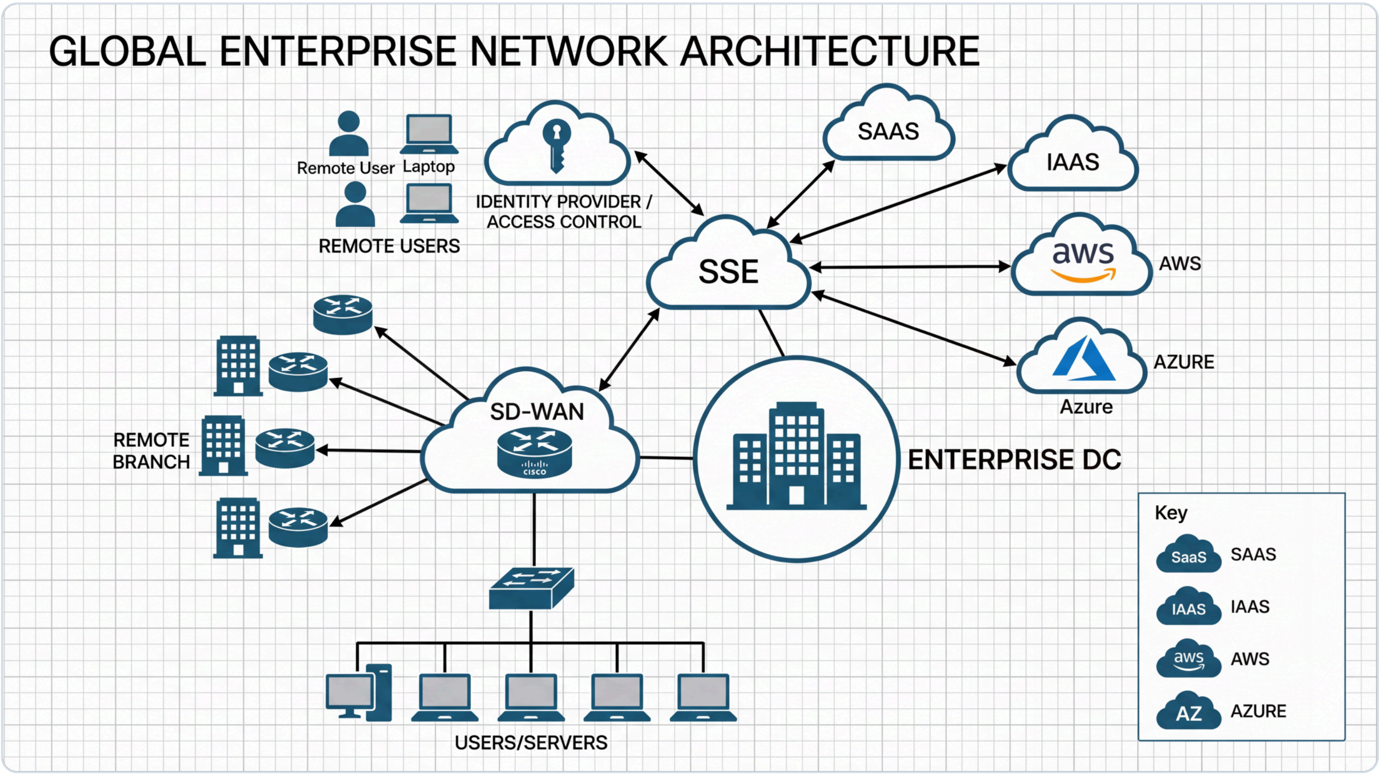
Scribe

- Responsible for documentation
- Final diagram and responses

Current Customer Architecture Diagram

The customer is a distributed enterprise: a corporate HQ and on-premises data center, a fleet of remote branches on SD-WAN, a growing remote and hybrid workforce, some of them now running AI agents, and an application estate spread across on-prem, AWS and Azure. They have grown by acquisition, they are mid-flight into public cloud, and their security controls have not kept pace with their footprint.

They are not asking you to rip anything out. They are asking you to bring Zero Trust to an environment that already exists, to segment it, to see it, and to keep enforcing policy no matter where the user, the device, or the workload happens to be.



Global enterprise network architecture: the customer high-level design (current environment).

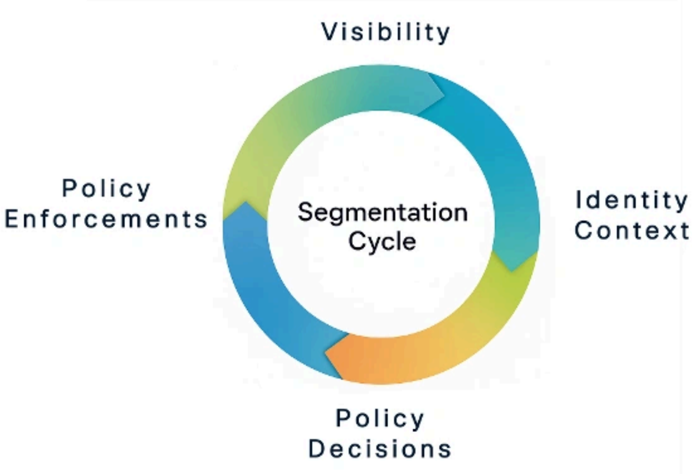
Group Tasks

- Use AI tools (i.e. Cisco Circuit app) to help you draw out your final diagram. For example, upload a rough sketch and have AI produce a nice professional network diagram as part of your proposal.
- Be able to explain to your SME your proposed design for each use case listed below and why you designed it your way. List any Cisco advantages or differentiators if available.

The Segmentation Cycle

A practical approach to network security. Apply this method in every use case

Cisco's practical approach to network security is an iterative loop, not a one-time project. Every use case in this clinic is a turn of this cycle: you gain visibility, add identity context, decide policy, then enforce it, and repeat as the environment changes.



How your work is scored

Each use case is worth **100 points**: **Mapping 40** (pains & requirements addressed) · **Products 30** (right Cisco products, how & why) · **Diagram 30**. Use cases 1-3 are required; 4-5 are bonus. Your **group** earns one level from the **cumulative** score across use cases, not per use case:

Associate

All three required use cases completed, cumulative score below 70% (under 210 of 300).

A working grasp of the fundamentals. The three required use cases are attempted with the right general direction, but the cumulative score is below 70%.

Professional

All three required use cases completed with a cumulative score of 70% or higher (210 of 300 or more). A perfect three-use-case result is Professional.

Solid, correct designs across the three required use cases: right products, correct integrations, and most requirements addressed. Attempting the two bonus use cases is encouraged but not required for this level.

Expert

All five use cases attempted with a cumulative score of 80% or higher (400 of 500 or more).

Mastery across the full brief: all five use cases are designed and the cumulative score is 80% or above, with Cisco differentiators articulated throughout.

Finish the three required use cases to be **Associate**, or **Professional** at 70% or higher across them (a perfect three-use-case result is Professional). To reach **Expert**, attempt all five use cases and score 80% or higher overall.

Cisco Zero Trust product primer

A quick reference to the products you might use. **You decide which apply** to each use case, and be ready to say how and why.

Cisco Identity Services Engine (ISE)

Authenticates and profiles every user and device on the network and assigns Security Group Tags (SGTs).

Cisco TrustSec / SGT

Segments traffic by an identity tag instead of VLANs or IP subnets; the policy follows the user or device anywhere.

Cisco Secure Access

Cloud security edge (SSE/SASE): Zero Trust Network Access, VPN-as-a-service, Secure Web Gateway, plus AI Access and an AI Gateway.

Cisco Secure Client

Endpoint agent with a full-VPN module and a Zero Trust Access (per-application) module.

Cisco Duo

Identity and access: multi-factor auth, Single Sign-On (Passport), passwordless, and Duo Directory as an identity provider (IdP).

Cisco Secure Firewall (FTD / FMC)

Next-generation firewall and IPS; can carry SGTs inline to extend segmentation between sites.

Cisco Secure Workload

Discovers applications (App Dependency Mapping) and enforces microsegmentation across on-prem, AWS and Azure.

Security Cloud Control (SCC)

One cloud console to manage Cisco security policy across products.

Cisco Catalyst SD-WAN
WAN fabric that integrates with Secure Access to deliver SASE to branches.

Isovalent (Cilium / eBPF / Tetragon)
Kubernetes networking, observability and runtime protection at the kernel level.

AI Access & AI Gateway (in Secure Access)
Discover and govern AI usage, enforce guardrails, and control which AI tools, MCP servers and agents are allowed.

Cisco AI Defense
Security and guardrails for AI applications and the data they touch.

Use Case #1: Managed Environment; Segmentation

Foundation · Required | 100 pts

CUSTOMER REQUIREMENTS

- Employee and trusted devices (Group=Employees)
- IoT (printers, cameras, badge readers, etc.) (Group=IoT)
- Deskside agents (dedicated Mac-minis) (Group=deskagents)
- Guest SSID for Internet only (Group=Guest)
- IoT should be restricted from the other groups but need data center and Internet access
- Guests should have Internet access only
- Employee and IoT require controlled internet, and data center application access
- Deskside Mac-minis have access to on-prem data center only

CUSTOMER PAIN POINTS

- Prior segmentation failed. Using VLANs/IP subnets is complex and tedious.
- Profiling is labor intensive and staff unable to keep up with constant new devices joining
- Failing PCI audits
- Corporate PCs not running latest AntiVirus and OS updates
- No budget for buying more firewalls to do segmentation

YOUR GROUP'S TASKS

- Draw a diagram of the customer's campus and branch wired/wireless networks. Submit for SME review and scoring.
- Create your proposed segmentation solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your proposed solution addresses each customer requirement and pain point.

YOUR PROPOSED DESIGN

Sketch your proposed network diagram here (or attach your Cisco Circuit / AI-generated diagram).
Record product choices, how each works, and the Cisco differentiator in your Workbook.

Use Case #2: Remote Worker: Zero Trust Access

Foundation · Required

100 pts

CUSTOMER REQUIREMENTS

- All corporate users can use full client VPN to access selected legacy private applications. But most prefer ZTNA for modern applications.
- All corporate users need controlled internet access (block gambling and gaming)
- Contractors can only access private applications via clientless RDP access.
- Improve user experience with minimal application sign-ons
- Customer wants SASE solution for branches

CUSTOMER PAIN POINTS

- Full client VPN has over permissive privileges. Not considered Zero Trust
- User satisfaction is low due to manual VPN connection daily
- Too many sign-ons per application. Low user satisfaction.
- Want to eliminate password logins to minimize stolen credentials
- Existing Okta IdP solution getting too expensive
- CIO wants SASE to reduce branch costs

YOUR GROUP'S TASKS

- Expand the diagram of the customer's campus and branch wired/wireless networks to support remote workers. Submit for SME review and scoring.
- Create your proposed secure remote worker solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your proposed solution addresses each customer requirement and pain point.

YOUR PROPOSED DESIGN

Sketch your proposed network diagram here (or attach your Cisco Circuit / AI-generated diagram).
Record product choices, how each works, and the Cisco differentiator in your Workbook.

Use Case #3: AI Access and Agent Controls

Foundation · Required

100 pts

CUSTOMER REQUIREMENTS

- User agents should not be granted rights to use tools equally. Fine grained control is needed.
- Corporate users should only be able to login to their organization's ChatGPT tenant
- Some remote users will be running Claude Desktop agents. Ensure your design includes controls for those employee systems.
- Some users will be integrating MCP servers to local agents like Claude Code

CUSTOMER PAIN POINTS

- Shadow AI usage is an issue today
- Too many unknown MCP servers for applications in both the private and public DCs.
- Users using personal credentials for non-enterprise ChatGPT login
- Downloading of risky AI models

YOUR GROUP'S TASKS

- Expand the diagram from your previous use cases, if necessary. Submit for SME review and scoring.
- Create your proposed agentic solution based on what was taught earlier in the agentic AI design presentation and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your proposed solution addresses each customer requirement and pain point.

YOUR PROPOSED DESIGN

Sketch your proposed network diagram here (or attach your Cisco Circuit / AI-generated diagram).
Record product choices, how each works, and the Cisco differentiator in your Workbook.

Use Case #4: Merger / Acquisition; Extend Segmentation

Bonus · If time permits | 100 pts

CUSTOMER REQUIREMENTS

- CISO wants to extend segmentation to acquired company using TrustSec.
- Only selected branch users can access a restricted resource at recently acquired company (Site 9951).

CUSTOMER PAIN POINTS

- Parent company is not ready to add 100 newly acquired staff into their corporate directory. Is there a quick option of using what they already have?
- During transition, minimize disruptive user authentication experience to corporate apps and quickly secure acquired company apps/user population
- Acquired company suffered identity breach previously. Still using passwords.

YOUR GROUP'S TASKS

- Expand the diagram from your previous use cases, if necessary. Submit for SME review and scoring.

- Create your proposed solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your proposed solution addresses each customer requirement and pain point.

YOUR PROPOSED DESIGN

Sketch your proposed network diagram here (or attach your Cisco Circuit / AI-generated diagram).
Record product choices, how each works, and the Cisco differentiator in your Workbook.

Use Case #5: Zero Trust across multicloud Data Centers

Bonus · If time permits | 100 pts

CUSTOMER REQUIREMENTS

- Limited budget. Want to leverage native security features when practical. Open to virtual appliances
- CIO seeking end-to-end segmentation solution
- Minimize complexity

CUSTOMER PAIN POINTS

- Minimal visibility of applications across multicloud data centers
- Lacking segmentation strategy and plan
- Wide deployment of Kubernetes with latency and visibility issues
- Unable to patch IoT devices
- Too many disparate security management interfaces for on-prem and multicloud enforcement points

YOUR GROUP'S TASKS

- Expand the diagram from your previous use cases, if necessary. Submit for SME review and scoring.
- Create your proposed solution and be ready to explain the product selected, how it works, and why (include Cisco differentiators if available). Make sure your proposed solution addresses each customer requirement and pain point.

YOUR PROPOSED DESIGN

Sketch your proposed network diagram here (or attach your Cisco Circuit / AI-generated diagram).
Record product choices, how each works, and the Cisco differentiator in your Workbook.

Glossary

ZTNA Zero Trust Network Access
Per-application access that replaces broad, over-permissive VPN.

SASE Secure Access Service Edge
Cloud-delivered networking and security combined.

SSE Security Service Edge
The security half of SASE (SWG, ZTNA, CASB, DLP…).

SGT Security Group Tag
An identity label used to segment traffic without VLANs or IP subnets.

TrustSec Cisco TrustSec
Cisco's SGT-based segmentation technology.

ISE Identity Services Engine
Authenticates and profiles users/devices and assigns SGTs.

SWG Secure Web Gateway
Inspects and filters internet-bound web traffic.

IdP Identity Provider
The system that authenticates users (e.g., Duo, Okta, Entra).

SSO Single Sign-On
One login for many applications.

MFA Multi-Factor Authentication
A second proof of identity beyond a password.

802.1X IEEE 802.1X
Port-based network access authentication.

MAB MAC Authentication Bypass
Authenticate a device by its MAC when it can't do 802.1X.

pxGrid Platform Exchange Grid
Cisco's framework for sharing context between security products.

SGACL Security Group ACL
Access rules written in terms of SGTs.

CSW Cisco Secure Workload
Application visibility and microsegmentation (formerly Tetration).

ADM Application Dependency Mapping
Discovers app traffic flows to build segmentation policy.

FTD / FMC Firewall Threat Defense / Firewall Mgmt Center
Cisco Secure Firewall and its manager.

NGFW / NGIPS Next-Gen Firewall / IPS
Modern firewall and intrusion-prevention.

MCP Model Context Protocol
How AI agents connect to external tools and data.

DCR Dynamic Client Registration
Registering and authorizing agents/apps with an IdP.

DLP Data Loss Prevention
Stops sensitive data from leaving the organization.

SCIM System for Cross-domain Identity Mgmt
Syncs users between directories (e.g., Okta → Duo).

VPNaaS VPN as a Service
Cloud-delivered VPN termination.

eBPF extended Berkeley Packet Filter
Kernel technology for observability and security (Cilium).

CNI Container Network Interface
Kubernetes networking plugin (e.g., Cilium).

SCC Security Cloud Control
Unified cloud management for Cisco security policy.

IoT Internet of Things
Networked devices like printers, cameras and badge readers.